KEDGE dart v29: build day run sheet

Dart: Ø42 x 152 mm, 1.5 mm walls, PLA on the X2D. Kick leg 0 to 201.75°, drill pair 0 to 42.25°. Collision sweep clean (kick in 203 steps to 201.75°, drills in 1° steps to 42.25°, every sub-part, exact meshes). Printed volume 89.67 cm³ (about 111.2 g if solid; less at 40% infill). About 12 hours wall clock with the two plates printing in the background. Rules for the whole day: the power loop plug stays OUT while you solder, move a wire or plug in USB. Never let the two cells touch metal together. One step at a time; each step says what you should see before you move on.

1. What is in the pack

File	What it is
print/plate_A_hull.3mf	Print plate A: hull top, hull bottom, nose, tail cap, both couplers. Already in print orientation.
print/plate_B_legs_tray.3mf	Print plate B: tray, kick leg, both drill pods, three augers. Already in print orientation.
print/fit_coupon.stl	20 minute test print: checks every fit before the long plates (section 4).
print/*.stl, step/*.step	Every printed part on its own, oriented for printing. STEP is the editable CAD (augers are faceted in STEP).
docs/wire_cut_list.csv	All 53 wires, pad to pad, colour, gauge and cut length (also section 9 of this sheet).
firmware/kedge_dart/	Firmware v3: kedge_dart.ino + kedge_logic.h. Keyboard over USB and WiFi, telemetry /status for the rover, current trip, auto-anchor (foot spins while the kick plants it, drills spin while the pods deploy), auto-on-landing. Compiled on ESP32 core 2.0.17 and 3.3.12.
firmware/test/	PC simulation of the control logic (all tests pass).
firmware/reel/reel_tension_limit	Reel sketch for the DevKitC + BTS7960: reel-in speed tapers and stops at a set tether tension (motor current or the load cell).
firmware/jetson/kedge_client.py	Rover / laptop client: status, logging, hold-a-key, auto, terminal driving, FPV overlay.
docs/KEDGE_*_engineering_report.pdf	Flight, launch and landing loads, holding force by soil, reel limit, launcher silhouette, design versions.
launcher/barrel_profile.dxf / barrel_keepout.stl	What the launcher barrel has to clear, with 1 mm clearance.
source/	The parametric generator and the whole check pipeline (run_all.sh). Every number lives in P at the top; KOV overrides make variants.

2. Parts and tools to have on the bench

Item	Qty	Note
XIAO ESP32S3, DRV8833 x3, INA219, BNO085, MP1584 (the 5.04 V one)	1 each	Headers clipped off (flush cutters), INA219 screw terminal cut off. MP1584 #2 stays spare.
OTOTEC 2S BMS stick	1	Caliper it when it arrives: the pocket under the cells is 40.8 x 8 x 2.1 plus 0.5 each way.
18350 cells, charged	2	Charge both to 4.2 V in the XTAR first.
Worm N20 (kick, deploy) + third worm N20 (foot, buy)	3	Buy the foot one single-shaft if listed; otherwise cut its lower stub to 2 mm.
Drill motors (12 x 10 x 24.27, M3 x 56 shaft)	2	
F693ZZ flanged bearings	5	2 per pod + 1 in the kick ear (bag has 10).
M3 nuts / M3 nylocs	2 / 2	Nut against each pod's inner bearing, nyloc in each auger.
M3 x 8 screws	7 + 2	7 countersunk (flat head, DIN 7991) for the radial hull screws, they sit flush with the skin; 2 pan head for the pod retainers (with washers).
M3 x 12 screw, M3 x 10 screw	1 + 1	x12 goes through the ear root into tray disc 1; x10 is the kick hub screw.
M3 washers	3	2 pod retainers + 1 between kick hub and ear bearing.
M3 grub screw	1	Foot auger on the foot N20 shaft flat.
2xAA battery holders (for their contacts)	2	You need 2 springs and 2 flat plates. Launch throws the cells backwards, so springs go at the FRONT of the cell bay and flat plates at the BACK.
470 to 1000 µF capacitor, 10 V or more	1	Across the MP1584 5 V output: rides through contact bounce on landing.
Schottky diode 1N5817 or SS14	1	In wire 18 (5 V to the XIAO): stops back-feeding USB.
Nose ballast 15 g		Steel shot or 3 to 4 M6 nuts set in 5-minute epoxy in the nose cavity. Without it the dart tumbles unless the tether is taut (engineering report, section 3).
JST-XH 3-pin pair, JST-XH 2-pin pair + a 2-pin shorting plug	1 each	Charge socket and power loop in the tail window.

Item	Qty	Note
Silicone wire 22, 24, 26, 28 AWG		Totals per colour in section 9.
Zip ties 2.5 mm, medium threadlocker (or CA), double-sided foam tape, hot glue		
Already on your bench		Caddx Ant camera, Zeus VTX + u.FL antenna, 3S LiPo + XT60 lead #1, XTAR MC2 (charges the 18350s out of the dart), breadboard and jumpers.
Tools		Soldering iron, solder, flux, multimeter, wire strippers, flush cutters, small flat screwdriver (MP1584 trimmer), heat shrink + lighter or heat gun, JST-XH crimper or pre-crimped pigtails, 5-minute epoxy, hex keys, tweezers, luggage scale (pull test).

3. Timeline

Clock	Step	Printer
0:00	4. Fit coupon	coupon (20 min)
0:30	5. Start plate A, flash the firmware	plate A (about 3 h)
1:00	6. Prepare boards, set both MP1584s, cut and label all wires	plate A
2:00	7. Solder the bench harness, power-up test, motor directions	plate A
3:30	Start plate B. Clean plate A parts, test-fit the hull, nose ballast (8a)	plate B (about 2.5 h)
4:00	8b. Battery contacts and BMS wiring off the dart	plate B
6:00	8c. Pods, couplers, augers (plate B done)	
7:00	8d. Populate the tray, battery power-up check	
8:30	8e. Close the hull, couplers, pods	
9:30	8f. Kick leg and foot	
10:15	10. Function test and calibration	
12:00	Done. Charge the cells (section 10, Charging).	

Bambu Studio settings (both plates): PLA, 0.4 nozzle, 0.16 mm layers, 4 wall loops (the 1.5 mm walls come out solid), 40% gyroid infill, top/bottom 5 layers, supports OFF, brim ON (auto) for the augers and couplers, X-Y hole compensation from the coupon (section 4). Plate times above are estimates; Studio shows the real ones.

Part	Print pose	Size on bed mm	Volume cm³
hull_top	split face down (on the bed), saddle up	121 x 50 x 35	17.3
hull_bottom	split face down, belly up	121 x 42 x 21	11.5
nose	tip down on the bed (flat Ø22), spigot up	42 x 42 x 27	8.25
tail_cap	end face down, spigot up (no supports)	42 x 42 x 15	7.13
tray	keel down on the bed; discs stand up, decks bridge 26 mm between discs	109 x 39 x 36	15.98
kick_leg	on its side (pole axis horizontal, hub face down)	119 x 15 x 15	8.7
drill_leg_L	upright, motor pocket open to the top (as installed folded)	46 x 15 x 13	3.37
drill_leg_R	upright, motor pocket open to the top (as installed folded)	46 x 15 x 13	3.37
drill_leg_L_auger	base down, nut slot sideways	16 x 16 x 71	4.63
drill_leg_R_auger	base down, nut slot sideways	16 x 16 x 71	4.63
foot_auger	base down	16 x 16 x 40	2.71
coupler_L	standing on its socket end	7 x 6 x 36	1.22
coupler_R	standing on its socket end	7 x 6 x 27	0.88

4. Fit coupon (0:00, 20 minutes)

Print fit_coupon.stl. It has, left to right: the worm N20 pocket, the drill motor pocket, the F693ZZ seat with its flange recess, the hull boss bore, the N20 D-bore, an M3 pilot, an M3 clearance hole, the pod D-bore, the 3.8 shaft clearance, a coupler peg, and a hull ring with its spigot ring.

Check	Should feel	If not
Worm N20 into the square pocket	Slides in by hand, no wobble	Tight: hole compensation +0.05. Loose: -0.05.
Drill motor into its pocket	Slides in by hand	Same as above
F693ZZ into the seat	Firm thumb press, flange flush in the recess	Too tight: +0.05. Falls out: a drop of CA on assembly day.
Coupler peg into the boss bore	Turns freely, little play	Tight: sand the peg lightly
Coupler peg D-socket onto an N20 shaft	Pushes on firmly, no rotation slop	Loose: a drop of CA on assembly
Coupler peg into the pod D-bore	Slides in, no rotation slop	
M3 screw into the 2.5 pilot	Cuts its own thread, holds	
Spigot ring into the hull ring	Push fit, no rattle	Loose: one layer of tape on the spigots

Directions used below: left = the kick-leg side, seen from the tail with the saddle on top (y plus in the CAD). The left pod is the brighter red one in the viewer.

5. Start plate A and flash the firmware (0:30)

1. Send plate_A_hull.3mf to the X2D.
2. In the Arduino IDE: board XIAO_ESP32S3, Tools > USB CDC On Boot: Enabled (the sketch will not compile otherwise). Library Manager: install Adafruit INA219 and Adafruit BNO08x.

3. Nothing else connected to the XIAO. Plug it in by USB-C, open firmware/kedge_dart/kedge_dart.ino, Upload. Re-select the board if the IDE switches it to the wrong one.
4. Serial Monitor at 115200, 'No line ending'. You should see the KEDGE ready help line (INA219 and BNO085 'not found' is normal now). If the monitor stays blank: press the XIAO reset button once, check the port and USB CDC On Boot.
5. On your phone or laptop, join WiFi KEDGE (password grappstar) and open 192.168.4.1: you get the control page. From now on you can also upload over WiFi: Arduino IDE > Port > the network port called kedge. Unplug USB.

6. Prepare boards and wires (1:00)

1. Clip every header pin flush on the XIAO and the three DRV8833 boards (flush cutters, close to the board). Cut the INA219 screw terminal off. Tin each pad you will use.
2. MP1584s, with the 3S LiPo on XT60 lead #1 and the multimeter on the output, turning the small brass screw with a small screwdriver: the dart one to 5.04 V; the spare one to 8.0 V (this is your bench motor supply in step 7). Disconnect.
3. Cut all wires to the lengths in section 9. Strip 3 mm each end, tin, and put a tape flag with the wire number on each. Always slide heat shrink onto a wire before you solder its second end.
4. DRV8833 check before any 3V3 wire: the driver chip has no logic supply pin. The pin called VIN (or VCC on some red boards) is the MOTOR supply. The 3V3 wires (25 to 27) go only to the SLP / nSLEEP / EEP pad, which wakes the driver. Meter on beep: touch the pad you plan to use and the VIN/VM pin. If it beeps, it is the motor supply: do not put 3V3 on it, or the battery voltage reaches the XIAO.
5. Diode: wire 18 has a Schottky diode (1N5817 or SS14) in it, stripe toward the XIAO, in heat shrink. It stops the 5 V buck back-feeding your Mac when USB is plugged in. Still never plug USB in while the dart is powered.

7. Solder the bench harness and test it (2:00)

Lay the boards on the bench in the order they sit in the dart (picture in section 11): sled boards (MP1584, DRV#3), XIAO + DRV#1, DRV#2 + INA219, BNO085. Solder in this order and check after each group:

1. Power: wires 10 to 13 and 15 to 24, and wire 14 at its MP1584 end only (its BMS end comes in 8b). Fit the capacitor across MP1584 OUT+ / OUT-: long leg to OUT+, the striped leg to OUT-. Snap the u.FL antenna onto the VTX now: never power a VTX without its antenna. Also solder wire 9 at its INA219 VIN+ end only (its socket end is crimped in 8b). Meter on beep: red rail to black rail must NOT beep.
2. 3V3 and ground to the logic: wires 25 to 31.
3. I2C: wires 32 to 35. Motor control: wires 36 to 43. Bridge AIN1 to BIN1 and AIN2 to BIN2 on DRV#2 with a short wire offcut.
4. Motor leads: solder wires 44 to 53 to the DRV outputs only; leave the far ends free.
5. Power-up test with the bench supply (the 8.0 V MP1584 on the LiPo): its OUT+ to the INA219 VIN+ pad, OUT- to the MP1584 IN- pad (main ground). You should see about 4.7 V on the XIAO 5V pad (5.04 minus the diode), the VTX LED on, and on the WiFi page: i reads about 8 V; z finds 0x40 and 0x4a. Disconnect the bench supply.
6. Motor directions, bench supply on again. Twist each motor lead temporarily onto its wire (no solder yet: the kick, drill and foot leads are threaded through holes later). Hold each key half a second and check:

Key	Motor	Must turn so that	If not
1	deploy N20	the output shaft turns the way that swings the pods DOWN (mark the shaft with a pen and compare with the viewer)	swap the pair on wire 44 / 45
2	kick N20	the hub would swing the leg UP over the nose	swap 46 / 47
3	drills	each drill output shaft turns: left motor clockwise seen from behind the motor, right motor anticlockwise (the augers are mirror images, so both then screw in)	one wrong: swap that motor's pair; both wrong: swap 40 / 41
4	foot N20	the foot N20 output shaft turns clockwise, seen from the shaft end (that screws the right-hand foot auger in)	swap 52 / 53

Note the free current on i for each (October bench: worms 20 to 30 mA, drill pair 60 to 70 mA). Mark each motor's lead that goes to the lower wire number. Solder the deploy N20 now (it never goes through a hole). Disconnect the bench supply.

8. Mechanical assembly

8a. Nose ballast (3:30, plate A done)

Stand the nose on its tip. Drop 15 g of steel shot or 3 to 4 M6 nuts into the cavity around the camera tube (not into the camera pocket) and cover them with 5-minute epoxy. Leave it to set while you carry on.

8b. Battery contacts and BMS, off the dart (4:00)

1. Join the two springs with wire 2 (the series link, 22 AWG) and solder wire 3 (BMS BM) to that link. Solder wire 1 to one flat plate and wire 4 to the other.
2. On the bench, solder wires 1, 3 and 4 to the BMS B-, BM and B+ pads, wire 14 to P-, wire 8 to P+, and wires 5 to 7 to B-, BM, B+ for the charge socket. Crimp or solder the JST-XH sockets on wires 5 to 7 (3-pin) and 8 / 9 (2-pin power loop).

8c. Drill pods, couplers and augers (6:00, plate B done)

1. Press one F693ZZ into the inner seat from the motor pocket side, flange toward the motor. Press the other into the nose end from outside, flange on the nose face.
2. Thread an M3 nut onto the drill motor shaft, about 3 mm from the motor face. Feed the shaft through both bearings, drop the motor into its pocket. Threadlocker on the shaft, then hold the nut with tweezers and turn the motor until the nut seats against the inner bearing. Zip-tie the motor in its groove.
3. Push an M3 nyloc into the side slot of the auger (nylon ring toward the point). Augers are handed: the LEFT pod takes drill_leg_L_auger (right-hand flight), the RIGHT pod takes drill_leg_R_auger (left-hand flight). Threadlocker on the shaft end, screw the auger on until its base boss touches the outer bearing.
4. Check: each auger spins freely with the motor, no rub on the pod nose.

8d. Populate the tray (7:00)

1. Glue the joined springs into the long recess on the FRONT cell-bay disc (nose side), one spring in front of each cell position. Glue the wire 1 plate into the BACK disc recess on the right (cell 1 side) and the wire 4 plate on the left. Lay the BMS in the channel under the cell bay.
2. Deploy N20 into the sled cradle (worm box toward the tail, output shaft across), zip tie through the groove under the cradle.
3. Stick the boards on with foam tape: DRV#3 on the keel and MP1584 on the shelf in the sled; BNO085 on deck A; XIAO (USB toward the nose) and DRV#1 on deck B; DRV#2 and INA219 on deck C; VTX on the deck over the cells. Route bundles through the notches in the discs. Antenna lead to the tail.

4. Push both sockets into the tail cap window, antenna through the top slot (wire 9 and wire 14 are already soldered from 7.1 and 8b). Tape over the charge socket now: its pins are live as soon as the cells go in.
5. Cells in: cell 1 (right) button toward the nose, cell 2 (left) button toward the tail. Each pushes in sideways, squeezing its spring, and sits hard against its back plate.
5a. Loop still out. Meter on the charge socket pins (lift the tape corner): pin 1 to 2 reads 3.6 to 4.2 V, pin 2 to 3 reads 3.6 to 4.2 V, pin 1 to 3 reads 7.2 to 8.4 V. Any negative reading: pull the cells out and check their direction.
6. Battery power-up check: check the u.FL antenna is still clicked onto the VTX, then plug the loop in. WiFi page: i reads 7.4 to 8.4 V; z finds 0x40 and 0x4a; the deploy N20 runs on 1. Pull the loop plug.

8e. Close the hull (8:30)

1. Kick N20 into the saddle on the hull top, worm box toward the nose, output shaft through both side holes. Zip tie: down one slot beside the saddle, under the skin, up the other, over the motor. Its leads go down through the hole in the saddle floor.
2. Camera into the nose pocket from the back (lens into the hole), cable out the back. Tray into the bottom half, and seat the nose spigot in the bottom half at the same time: the groove on the spigot sits over the half-rib inside the hull.
3. Thread the leads before the top half goes on: kick leads (wires 46, 47) up through the saddle floor hole; foot leads (52, 53) out through the skin hole beside the saddle; drill leads (48 to 51) out through the two flank holes. Lower the top half on.
4. Radial screws into disc 1: M3 x 8 at left z -3, right z +3 and right z -3, and the M3 x 12 through the ear root at left z +3; four M3 x 8 into the tail spigot (z ±3 both sides). The top half's rib closes the nose groove, so the nose is locked.
5. Solder the kick N20 and the two drill motors to their leads (as marked in 7.6), heat shrink (slide it on before soldering; iron on each joint 3 s at most; shrink with a heat gun on low, no lighter near the PLA).
6. Couplers: slide coupler_L in through the left boss, coupler_R through the right, turning gently until each D-socket drops onto the deploy N20 stub. Pods onto the coupler ends (D-bore), then M3 x 8 + washer into each coupler end. Fold the pods along the flanks.

8f. Kick leg and foot (9:30)

1. Press an F693ZZ into the ear from outside, flange in the recess.
2. Foot N20 into the foot housing (opening on the inboard side), output shaft up through the top hole, zip tie. Foot auger onto the shaft, grub screw onto the flat.
3. Feed the foot leads from the foot pocket through the pole channel and out the small hole near the hub. Solder them to wires 52 and 53 (as marked in 7.6): heat shrink on first, iron 3 s at most per joint, heat gun on low, no lighter near the PLA. Leave the loop round the hub.
4. Push the hub onto the kick N20 stub (D flat), put the washer between hub and ear, M3 x 10 through the ear bearing into the hub. Plug the loop in and send @ until the leg rests folded on the hull.

10. Function test and calibration (10:15)

Use a tray of damp soil or sand. Laptop on the KEDGE WiFi page; hold a key to run, let go to stop. Space bar stops everything.

Key	Should happen	If it goes the wrong way
First, in the air	Hold each key for half a second: directions were fixed in 7.6	If one is still wrong, swap it at the DRV end (open the hull)
2 (hold)	Kick leg swings up over the nose and down; the foot auger tip meets the soil near 180°	Swap wires 46 and 47
2 and 4 (hold both)	Foot auger screws IN while the kick keeps pushing, until the foot housing sits on the soil	Foot backs out: swap wires 52 and 53
1 and 3 (hold both)	Both pods swing down and back while BOTH augers spin and cut in; the dart does not twist	Pods go up: swap 44 and 45. One auger backs out: swap that motor's pair (48/49 or 50/51). Both: swap 40 and 41 at the XIAO
i during each	Note the free current, the current when the foot touches, and the stall current	Put them in kedge_logic.h (kickContactMa, kickStallMa, deployStallMa) and upload over WiFi
b	Roll and pitch of the planted dart	
a	Full auto sequence, same order as above, by itself	Space aborts
A then throw	Arms auto-anchor on landing (the dart anchors 1.5 s after it comes to rest)	Press A again to disarm

Charging: either take the cells out and use the XTAR MC2, or plug a 2S balance charger (JST-XH 3-pin) into the charge socket in the tail. Loop plug OUT while charging.
Recovery: # (drills out), ! (pods up), $ (foot out), @ (kick back). The worms hold every position with the power off.
Note: STOP (space, /stop, or a client hold ending) also aborts a running auto sequence; that is deliberate.
Pin change from the October bench: the drill pair is now on D8/D9 and the foot on D6/D7. D6/D7 are the XIAO's serial pins and flicker at every reset; the slow foot worm does not care, the drills would twitch.

9. Wire cut list (pad to pad)

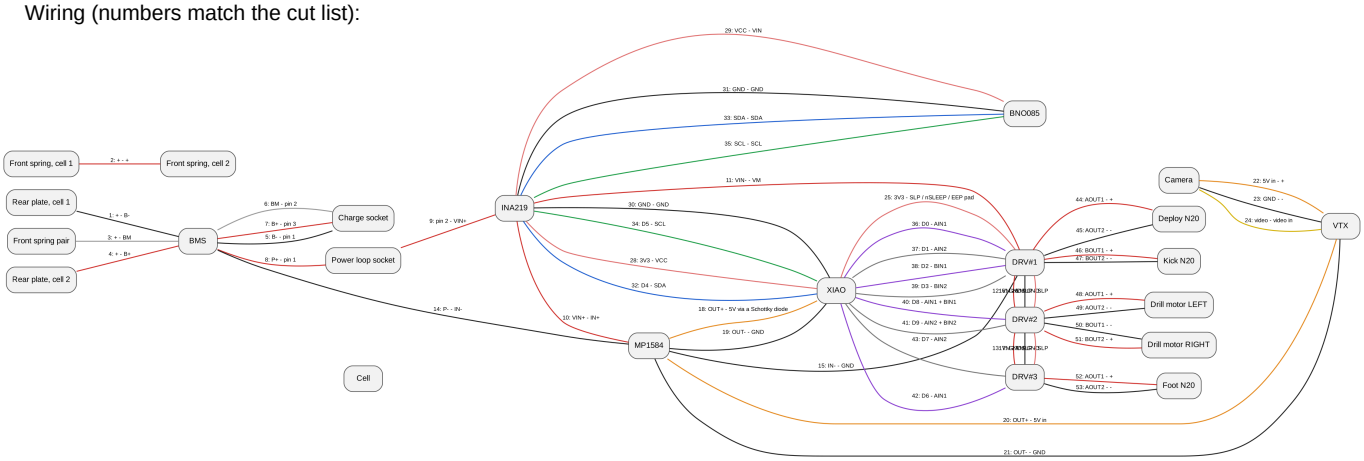
#	From	To	Colour	Gauge	Cut mm	Note
1	Rear plate, cell 1 (-)	BMS B-	black	22 AWG	45	solder to the plate tab before it goes in its recess
2	Front spring, cell 1 (+)	Front spring, cell 2 (-)	red	22 AWG	45	the series link: carries the full battery current; glue both springs in the front recess
3	Front spring pair (cell 1 + / cell 2 -)	BMS BM	white	26 AWG	55	the two springs are joined by a short wire; this is the midpoint sense
4	Rear plate, cell 2 (+)	BMS B+	red	22 AWG	45	
5	BMS B-	Charge socket pin 1 (tail window)	black	26 AWG	80	JST-XH 3-pin, balance charge
6	BMS BM	Charge socket pin 2	white	26 AWG	125	
7	BMS B+	Charge socket pin 3	red	26 AWG	80	
8	BMS P+	Power loop socket pin 1 (tail window)	red	22 AWG	80	JST-XH 2-pin; the shorting plug is the ON switch
9	Power loop socket pin 2	INA219 VIN+	red	22 AWG	95	main + after the loop
10	INA219 VIN+ (same pad)	MP1584 IN+	red	22 AWG	140	buck taps before the shunt, so INA219 reads motor current only
11	INA219 VIN-	DRV#1 VM	red	22 AWG	50	motor rail 6.0-8.4 V
12	DRV#1 VM	DRV#2 VM	red	22 AWG	65	daisy chain
13	DRV#2 VM	DRV#3 VM	red	22 AWG	155	forward into the sled (keel)
14	BMS P-	MP1584 IN-	black	22 AWG	130	main ground
15	MP1584 IN- (same pad)	DRV#1 GND	black	22 AWG	120	
16	DRV#1 GND	DRV#2 GND	black	22 AWG	65	
17	DRV#2 GND	DRV#3 GND	black	22 AWG	155	
18	MP1584 OUT+ (5.04 V)	XIAO 5V via a Schottky diode (stripe toward the XIAO)	orange	24 AWG	160	set and check 5.04 V BEFORE connecting; the diode stops the buck back-feeding USB (1N5817 or SS14, in heat shrink)
19	MP1584 OUT-	XIAO GND	black	24 AWG	160	
20	MP1584 OUT+	VTX 5V in	orange	26 AWG	95	
21	MP1584 OUT-	VTX GND	black	26 AWG	95	
22	VTX 5V in (same pad)	Camera + (red)	orange	28 AWG	105	runs forward along the top through the disc notches
23	VTX GND (same pad)	Camera - (black)	black	28 AWG	105	
24	Camera video (yellow)	VTX video in	yellow	28 AWG	105	twist with the camera ground
25	XIAO 3V3	DRV#1 SLP / nSLEEP / EEP pad (NOT VCC or VIN)	red/white	28 AWG	55	wakes the driver. Check first: if the pad you picked beeps to VIN/VM on the meter, it is the motor supply: do NOT connect 3V3 to it
26	DRV#1 SLP	DRV#2 SLP	red/white	28 AWG	65	
27	DRV#2 SLP	DRV#3 SLP	red/white	28 AWG	155	
28	XIAO 3V3	INA219 VCC	red/white	28 AWG	60	
29	INA219 VCC	BNO085 VIN	red/white	28 AWG	60	
30	XIAO GND	INA219 GND	black	28 AWG	60	
31	INA219 GND	BNO085 GND	black	28 AWG	60	
32	XIAO D4	INA219 SDA	blue	28 AWG	60	I2C SDA
33	INA219 SDA	BNO085 SDA	blue	28 AWG	60	
34	XIAO D5	INA219 SCL	green	28 AWG	60	I2C SCL
35	INA219 SCL	BNO085 SCL	green	28 AWG	60	
36	XIAO D0	DRV#1 AIN1	purple	28 AWG	55	deploy
37	XIAO D1	DRV#1 AIN2	grey	28 AWG	55	deploy
38	XIAO D2	DRV#1 BIN1	purple	28 AWG	55	kick
39	XIAO D3	DRV#1 BIN2	grey	28 AWG	55	kick
40	XIAO D8	DRV#2 AIN1 + BIN1 (bridge the two pads)	purple	28 AWG	45	drills (moved off D6/D7: those twitch at reset)
41	XIAO D9	DRV#2 AIN2 + BIN2 (bridge the two pads)	grey	28 AWG	45	drills
42	XIAO D6	DRV#3 AIN1	purple	28 AWG	145	foot (slow worm, harmless if it twitches at reset)
43	XIAO D7	DRV#3 AIN2	grey	28 AWG	145	foot

#	From	To	Colour	Gauge	Cut mm	Note
44	DRV#1 AOUT1	Deploy N20 + (red)	red	26 AWG	165	motor sits in the sled cradle
45	DRV#1 AOUT2	Deploy N20 - (black)	black	26 AWG	165	
46	DRV#1 BOUT1	Kick N20 + (red)	red	26 AWG	150	out through the hole in the saddle floor
47	DRV#1 BOUT2	Kick N20 - (black)	black	26 AWG	150	
48	DRV#2 AOUT1	Drill motor LEFT + (red)	red	26 AWG	195	out the left flank hole, 30 mm loop for the pod swing
49	DRV#2 AOUT2	Drill motor LEFT - (black)	black	26 AWG	195	
50	DRV#2 BOUT1	Drill motor RIGHT - (black) SWAPPED	black	26 AWG	170	reversed on purpose: the pair counter-rotates
51	DRV#2 BOUT2	Drill motor RIGHT + (red) SWAPPED	red	26 AWG	170	
52	DRV#3 AOUT1	Foot N20 + (red)	red	26 AWG	255	out the skin hole, 60 mm loop round the hub, through the pole channel
53	DRV#3 AOUT2	Foot N20 - (black)	black	26 AWG	255	silicone wire, it flexes every deploy

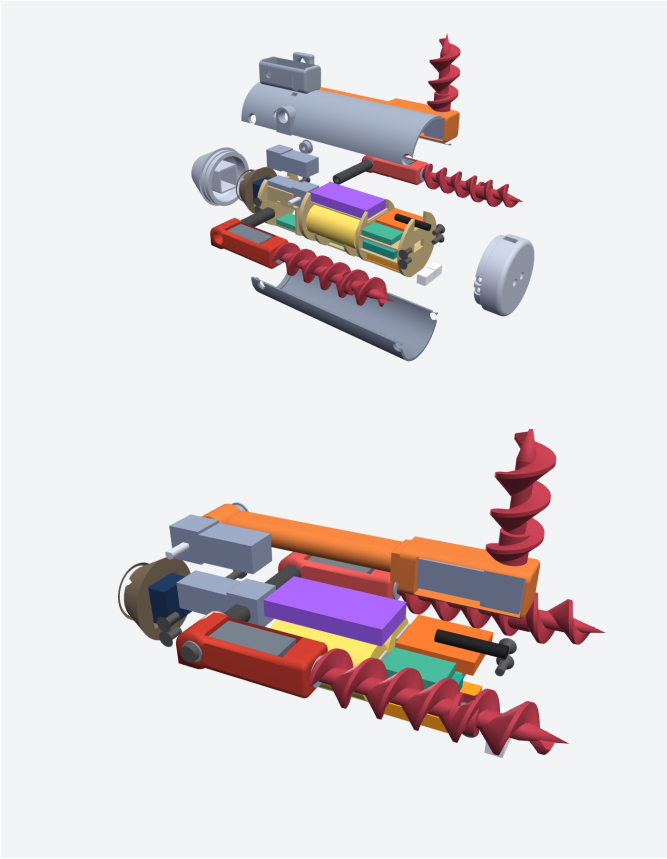
Totals by gauge and colour (mm): 22 AWG black 515, 22 AWG red 675, 24 AWG orange 160, 24 AWG black 160, 26 AWG white 180, 26 AWG black 1110, 26 AWG red 1015, 26 AWG orange 95, 28 AWG orange 105, 28 AWG black 225, 28 AWG yellow 105, 28 AWG red/white 395, 28 AWG blue 120, 28 AWG green 120, 28 AWG purple 300, 28 AWG grey 300

11. Pictures

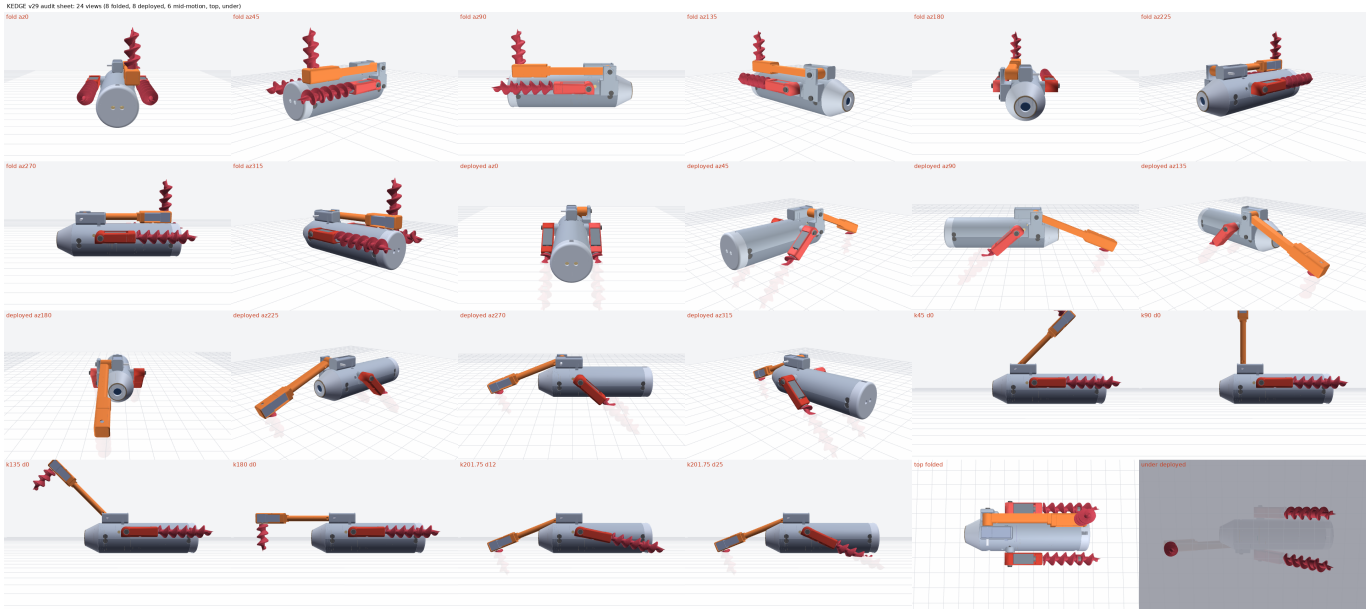
Wiring (numbers match the cut list):



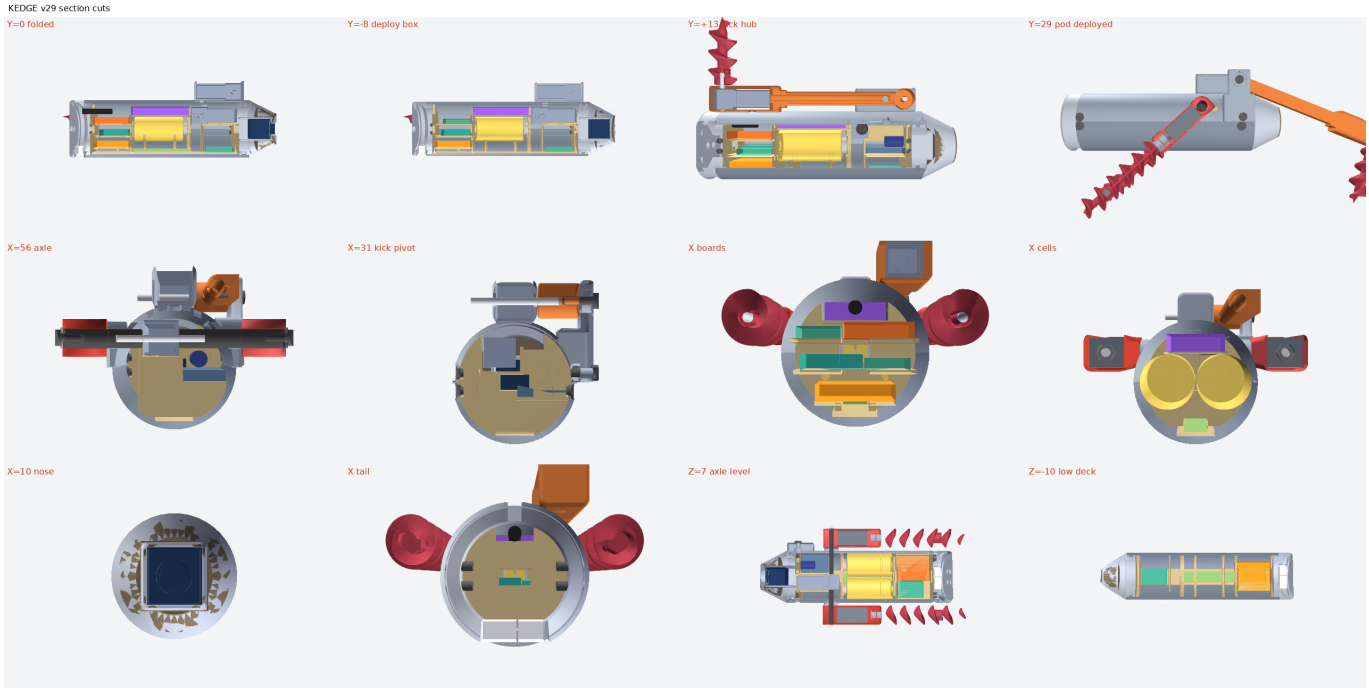
Exploded view and populated tray:



Audit sheet: 24 views of the exact print geometry (8 folded, 8 deployed, 6 mid-motion, top, under).



Section cuts:



12. Design notes

• Pod thrust: the M3 nut against the inner bearing takes the auger's pull into the ground; the auger base against the outer bearing takes push. The gearbox only carries torque.
• The drill pair counter-rotates so the dart does not spin; the two augers are mirror images so both screw in.
• The deploy N20's own shaft stubs drive the pods through two printed Ø7 D couplers that run in the hull bosses.
• Tether load path: tail end wall, tail cap, 4 radial M3 into the hull. Tie the line through the two end holes round the web.
• Ground geometry: kick stops at 201.75° with the foot housing on the ground and the foot auger 35.5 mm in; drills stop at 42.25° with the pods 0.52 mm clear and the augers 43.4 mm in.
• Power: 2S straight to the DRV8833 motor rail through the INA219; one MP1584 at 5.04 V for the XIAO, VTX and camera.
• Below the centreline the hull is a plain Ø42 cylinder for the launcher cradle; the folded pods sit just above it.